6.7 Continuous-Thrust Transfers

We will study continuous thrust transfers in the class for optimal control if you join the graduate study.

6.8 Relative Motion

Rendezvous in space

Satellite formation flying

- Motion between two close-orbiting satellites
  - In the frame of Two-Body Problem
  - Proximity often justifies analysis by linearization

  - Hill (=Clohessy-Withshire (CW) = Hill-Clohessy-Wiltshire (HCW)) equation describes linear motion with respect to a (passive) target spacecraft in Circular reference orbit

  - ➢ Tschauner-Hempel equation describes linear motion with respect to a (passive) target spacecraft in Elliptic reference orbit

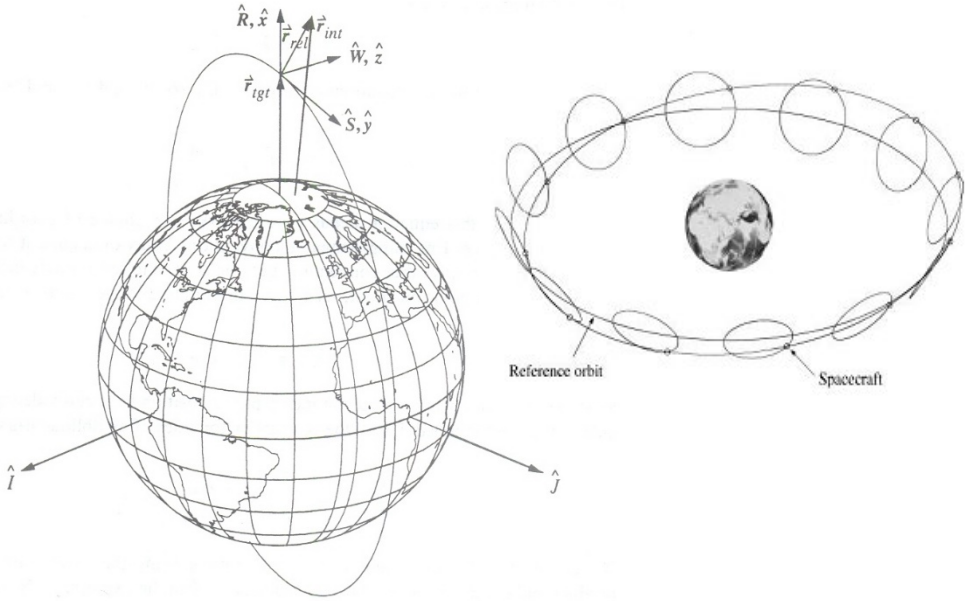


Figure 6-25. Coordinate-System Geometry for Relative Motion. The *xyz* notation reminds us the results are approximate. Remember that the *S* (and *y*) axes will be aligned with the velocity vector only in circular orbits. A major assumption in relative motion is that the target and interceptor satellites are in nearly circular orbits.

- Derivation of the Hill equation

  Start by defining the RSW coordinate system

  R: collinear with the position vector

  S: aligned with the local horizontal along the velocity vector

  W: normal to the orbital plane.

  Define the equations of motion for the target satellite in the frame of two-body equations of motion.

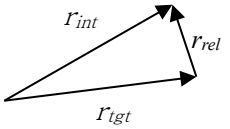
  Inertial acceleration:

$$\ddot{\vec{r}}_{tgt} = -\frac{\mu}{r_{tgt}^3}\vec{r}_{tgt} \qquad (6-46)$$

$$\ddot{\vec{r}}_{int} = -\frac{\mu}{r_{int}^3}\vec{r}_{int} + \vec{F}_{thrust} \qquad (6-47)$$

$$\vec{r}_{rel} = \vec{r}_{int} - \vec{r}_{tgt}$$

$$\ddot{\vec{r}}_{rel} = \ddot{\vec{r}}_{int} - \ddot{\vec{r}}_{tgt} = -\frac{\mu}{r_{int}^3}\vec{r}_{int} + \vec{F}_{thrust} + \frac{\mu}{r_{tgt}^3}\vec{r}_{tgt} \qquad (6-48)$$



$$\vec{r}_{int}\cdot\vec{r}_{int} = (\vec{r}_{tgt}+\vec{r}_{rel})\cdot(\vec{r}_{tgt}+\vec{r}_{rel})$$
$$r_{int}^2 = r_{tgt}^2 + 2\vec{r}_{tgt}\cdot\vec{r}_{rel} + r_{rel}^2$$

$$\frac{\vec{r}_{int}}{r_{int}^3} = \frac{\vec{r}_{tgt}+\vec{r}_{rel}}{(r_{tgt}^2 + 2\vec{r}_{tgt}\cdot\vec{r}_{rel} + r_{rel}^2)^{3/2}}$$

$$= \frac{\left(\vec{r}_{tgt}+\vec{r}_{rel}\right)}{r_{tgt}^3}\left\{\frac{1}{\left(1+\frac{2\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}^2}+\cancel{\frac{r_{rel}^2}{r_{tgt}^2}}\right)^{3/2}}\right\} \qquad \Leftarrow \quad r_{rel} << r_{tgt}$$

$$\approx \frac{\left(\vec{r}_{tgt}+\vec{r}_{rel}\right)}{r_{tgt}^3}\left(1+\frac{2\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}^2}\right)^{-3/2} \qquad \Leftarrow \quad (1+x)^n = 1+nx+\frac{n(n-1)x^2}{2!}+\cdots \text{ (binomial expansion)}$$

$$\approx \frac{\left(\vec{r}_{tgt}+\vec{r}_{rel}\right)}{r_{tgt}^3}\left\{1-\frac{3}{2}\frac{2\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}^2}\right\}$$

Substituting this result into (6-48) results in

$$\ddot{\vec{r}}_{rel} = -\frac{\mu}{r_{tgt}^3}\left[\cancel{\vec{r}_{tgt}}+\vec{r}_{rel}-3\vec{r}_{tgt}\left(\frac{\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}^2}\right)-3\vec{r}_{rel}\left(\frac{\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}^2}\right)\right]+\vec{F}_{thrust}+\cancel{\frac{\mu}{r_{tgt}^3}\vec{r}_{tgt}}$$

$$= -\frac{\mu}{r_{tgt}^3}\left[\vec{r}_{rel}-3\vec{r}_{tgt}\left(\frac{\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}^2}\right)-3\vec{r}_{rel}\cancel{\left(\frac{\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}^2}\right)}\right]+\vec{F}_{thrust} \qquad \Leftarrow r_{rel} << r_{tgt}$$

$$\approx -\frac{\mu}{r_{tgt}^3}\left[\vec{r}_{rel}-3\frac{\vec{r}_{tgt}}{r_{tgt}}\left(\frac{\vec{r}_{tgt}\cdot\vec{r}_{rel}}{r_{tgt}}\right)\right]+\vec{F}_{thrust}, \qquad \Leftarrow \frac{\vec{r}_{tgt}}{r_{tgt}} = \hat{R},\ \hat{R}\cdot\vec{r}_{rel} = x$$

$$= -\frac{\mu}{r_{tgt}^3}\left[-3x\hat{R}+\vec{r}_{rel}\right]+\vec{F}_{thrust} \qquad (6-49)$$

(6-49) represents the <u>inertial</u> acceleration of the interceptor in the target frame
However, as we study relative motion of the interceptor with respect to a target, it is desired to describe the value of $\ddot{\vec{r}}_{rel}$ in the <u>rotating coordinate frame</u>

Recall the relationship between inertial and relative vectors in a rotating frame:

$$\dot{\vec{r}}_I = \dot{\vec{r}}_R + \vec{\omega}_R \times \vec{r}_R$$

$$\begin{aligned}
\ddot{\vec{r}}_I &= \overset{I}{\frac{d}{dt}}\left(\dot{\vec{r}}_I\right) \\
&= \overset{R}{\frac{d}{dt}}\left(\dot{\vec{r}}_I\right) + \vec{\omega}_R \times \dot{\vec{r}}_I \\
&= \overset{R}{\frac{d}{dt}}\left(\dot{\vec{r}}_R + \vec{\omega}_R \times \vec{r}_R\right) + \vec{\omega}_R \times \left(\dot{\vec{r}}_R + \vec{\omega}_R \times \vec{r}_R\right) \\
&= \ddot{\vec{r}}_R + \dot{\vec{\omega}}_R \times \vec{r}_R + 2\vec{\omega}_R \times \dot{\vec{r}}_R + \vec{\omega}_R \times \left(\vec{\omega}_R \times \vec{r}_R\right)
\end{aligned}$$

$$\Rightarrow \ddot{\vec{r}}_R = \ddot{\vec{r}}_I - \dot{\vec{\omega}}_R \times \vec{r}_R - 2\vec{\omega}_R \times \dot{\vec{r}}_R - \vec{\omega}_R \times \left(\vec{\omega}_R \times \vec{r}_R\right), \quad \omega_R = \sqrt{\frac{\mu}{r_{tgt}^3}}, \quad \dot{\vec{\omega}}_R = 0 \text{ for circular orbit}$$

ω_R is the mean motion of target s/c (average angular motion over one orbital period)

$$\ddot{\vec{r}}_R \Rightarrow \ddot{\vec{r}}_{rel_R}, \quad \ddot{\vec{r}}_I \Rightarrow \ddot{\vec{r}}_{rel_I}, \quad \vec{r}_R \Rightarrow \vec{r}_{rel_R}$$

$$\begin{aligned}
\ddot{\vec{r}}_{rel_R} &= \ddot{\vec{r}}_{rel_I} - \cancel{\dot{\vec{\omega}}_R} \times \vec{r}_{rel_R} - 2\vec{\omega}_R \times \dot{\vec{r}}_{rel_R} - \vec{\omega}_R \times \left(\vec{\omega}_R \times \vec{r}_{rel_R}\right) \\
&= \ddot{\vec{r}}_{rel_I} - 2\vec{\omega}_R \times \dot{\vec{r}}_{rel_R} - \vec{\omega}_R \times \left(\vec{\omega}_R \times \vec{r}_{rel_R}\right) \qquad (6-50)
\end{aligned}$$

$$\vec{\omega}_R \times \dot{\vec{r}}_{rel_R} = \begin{vmatrix} \hat{R} & \hat{S} & \hat{W} \\ 0 & 0 & \omega \\ \dot{x} & \dot{y} & \dot{z} \end{vmatrix} = -\omega\dot{y}\hat{R} + \omega\dot{x}\hat{S}$$

$$\vec{\omega}_R \times \left(\vec{\omega}_R \times \vec{r}_{rel_R}\right) = \vec{\omega}_R \times \begin{vmatrix} \hat{R} & \hat{S} & \hat{W} \\ 0 & 0 & \omega \\ x & y & z \end{vmatrix} = \vec{\omega}_R \times \left(-\omega y\hat{R} + \omega x\hat{S}\right)$$

$$= \begin{vmatrix} \hat{R} & \hat{S} & \hat{W} \\ 0 & 0 & \omega \\ -\omega y & \omega x & 0 \end{vmatrix} = -\omega^2 x\hat{R} - \omega^2 y\hat{S}$$

Substituting these relationships and (6-49) into (6-50) yields

$$\ddot{\vec{r}}_{rel_R} = -\frac{\mu}{r_{tgt}^3}\left[-3x\hat{R} + \vec{r}_{rel_R}\right] + \vec{F}_{thrust} - 2\left(-\omega\dot{y}\hat{R} + \omega\dot{x}\hat{S}\right) + \omega^2 x\hat{R} + \omega^2 y\hat{S}$$

where $\vec{r}_{rel_R} = x\hat{R} + y\hat{S} + z\hat{W}$, $\ddot{\vec{r}}_{rel_R} = \ddot{x}\hat{R} + \ddot{y}\hat{S} + \ddot{z}\hat{W}$, $\omega^2 = \frac{\mu}{r_{tgt}^3}$, $\vec{F}_{thrust} = f_x\hat{R} + f_y\hat{S} + f_z\hat{W}$

$$\Rightarrow \begin{cases} \hat{R} \text{ component}: & \ddot{x} = 3\omega^2 x - \omega^2 x + 2\omega\dot{y} + \omega^2 x + f_x \\ \hat{S} \text{ component}: & \ddot{y} = -\omega^2 y - 2\omega\dot{x} + \omega^2 y + f_y \\ \hat{W} \text{ component}: & \ddot{z} = -\omega^2 z + f_z \end{cases}$$

$$\Rightarrow \begin{cases} \ddot{x} - 2\omega\dot{y} - 3\omega^2 x = f_x \\ \ddot{y} + 2\omega\dot{x} = f_y \\ \ddot{z} + \omega^2 z = f_z \end{cases} \qquad \underline{\text{Hill (CW; HCW) equations}} \qquad (6\text{-}51)$$

Note again that the Hill equations are represented in the <u>rotating</u> coordinate system

6.8.1 Position Solutions

- Solution to the Hill equation under no external forces of the interceptor $\left(\vec{F}_{thrust}=0\right)$
  - Radial Component

$$\ddot{x}=3\omega^2 x+2\omega\dot{y}$$
$$\Rightarrow \dddot{x}=3\omega^2\dot{x}+2\omega\ddot{y} \quad \leftarrow \quad \ddot{y}=-2\omega\dot{x}$$
$$\Rightarrow \dddot{x}+\omega^2\dot{x}=0$$

Take Laplace transform

$$L\left[\dddot{x}+\omega^2\dot{x}\right]=\left\{s^3X(s)-s^2x_0-s\dot{x}_0-\ddot{x}_0\right\}+\omega^2\left\{sX(s)-x_0\right\}=0$$

$$sX(s)(s^2+\omega^2)=(s^2+\omega^2)x_0+s\dot{x}_0+\ddot{x}_0$$

$$\Rightarrow X(s)=\frac{x_0}{s}+\frac{\dot{x}_0}{(s^2+\omega^2)}+\frac{\ddot{x}_0}{s(s^2+\omega^2)}$$

$$\begin{cases} \dfrac{\ddot{x}_0}{s(s^2+\omega^2)}=\dfrac{A}{s}+\dfrac{Bs+C}{(s^2+\omega^2)} \\ \qquad\qquad =\dfrac{As^2+A\omega^2+Bs^2+Cs}{s(s^2+\omega^2)} \\ \qquad\qquad =\dfrac{(A+B)s^2+Cs+A\omega^2}{s(s^2+\omega^2)} \\ \rightarrow (A+B)=0,\quad C=0,\quad A\omega^2=\ddot{x}_0 \\ \rightarrow \quad A=\dfrac{\ddot{x}_0}{\omega^2},\quad C=0,\quad B=-\dfrac{\ddot{x}_0}{\omega^2} \end{cases}$$

$$\Rightarrow X(s)=\frac{x_0}{s}+\frac{\dot{x}_0}{(s^2+\omega^2)}+\frac{\ddot{x}_0}{s\omega^2}-\frac{s\ddot{x}_0}{\omega^2(s^2+\omega^2)}$$

Take the inverse Laplace transform

$$x(t)=x_0+\frac{\dot{x}_0}{\omega}\sin(\omega t)+\frac{\ddot{x}_0}{\omega^2}-\frac{\ddot{x}_0}{\omega^2}\cos(\omega t)$$

$$(\quad \ddot{x}_0=3\omega^2x_0+2\omega\dot{y}_0)$$

$$\Rightarrow x(t)=4x_0+\frac{2\dot{y}_0}{\omega}+\frac{\dot{x}_0}{\omega}\sin(\omega t)-(3x_0+\frac{2\dot{y}_0}{\omega})\cos(\omega t)$$

$$\Rightarrow \dot{x}(t)=\frac{\dot{x}_0}{\omega}\omega\cos(\omega t)+(3x_0+\frac{2\dot{y}_0}{\omega})\omega\sin(\omega t)$$

$$=\dot{x}_0\cos(\omega t)+(3x_0\omega+2\dot{y}_0)\sin(\omega t)$$

- Tangential Component

Substitute $\dot{x}(t)$ into the $\ddot{y}$ equation of the Hill equations

$$\ddot{y}+2\omega\dot{x}=0$$

$$\ddot{y}=-2\omega\left(\dot{x}_0\cos(\omega t)+(3x_0\omega+2\dot{y}_0)\sin(\omega t)\right)$$

$$=-2\omega\dot{x}_0\cos(\omega t)-2\omega(3x_0\omega+2\dot{y}_0)\sin(\omega t)$$

Integrate the equation to find $\dot{y}$ and y:

$$\dot{y}=-2\dot{x}_0\sin(\omega t)+2(3x_0\omega+2\dot{y}_0)\cos(\omega t)+C$$

$$y=\frac{2\dot{x}_0}{\omega}\cos(\omega t)+\left(6x_0+\frac{4\dot{y}_0}{\omega}\right)\sin(\omega t)+Ct+D$$

To determine C and D, substitute t=0 into $\dot{y}$ and y

$$\dot{y}_0=6x_0\omega+4\dot{y}_0+C \quad \Rightarrow \quad C=-6x_0\omega-3\dot{y}_0$$

$$y_0=\frac{2\dot{x}_0}{\omega}+D \quad \Rightarrow \quad D=-\frac{2\dot{x}_0}{\omega}+y_0$$

Then, the answer is

$$y(t)=\frac{2\dot{x}_0}{\omega}cos(\omega t)+\left(6x_0+\frac{4\dot{y}_0}{\omega}\right)sin(\omega t)-(6\omega x_0+3\dot{y}_0)t-\frac{2\dot{x}_0}{\omega}+y_0$$

- Normal Component

  For z equation, we have a simple harmonic oscillator

$$\ddot{z}+\omega^2 z=0$$

  Take Laplace transform

$$s^2Z(s)-sz_0-\dot{z}_0+\omega^2Z(s)=0$$

$$Z(s)(s^2+\omega^2)=sz_0+\dot{z}_0$$

$$Z(s)=\frac{sz_0}{(s^2+\omega^2)}+\frac{\dot{z}_0}{(s^2+\omega^2)}$$

  The solution in the time domain is

$$z(t)=z_0cos(\omega t)+\frac{\dot{z}_0}{\omega}sin(\omega t)$$

- The resulting motion is somewhat elliptical

  Although the interceptor does not actually orbit the target S/C, the instantaneous parameters often result in an elliptical-like motion

- Differentiate the position equations to obtain the velocity with respect to the target:

$$\dot{x}(t)=\dot{x}_0cos(\omega t)+(3x_0\omega+2\dot{y}_0)sin(\omega t)$$

$$\dot{y}(t)=-2\dot{x}_0sin(\omega t)+\left(6\omega x_0+4\dot{y}_0\right)cos(\omega t)-(6\omega x_0+3\dot{y}_0)$$

$$\dot{z}(t)=-z_0\omega sin(\omega t)+\dot{z}_0cos(\omega t)$$

<Table of Laplace Transformation>

F(s)	*f(t)*
1	Unit impulse $\delta(t)$
$\frac{1}{s}$	Unit step 1(t)
$\frac{1}{s^2}$	t
$\frac{1}{s^n}$	$\frac{t^n}{(n-1)!}$ (n = 1, 2, 3…)
$\frac{n!}{s^{n+1}}$	t^n (n = 1, 2, 3…)
$\frac{1}{\sqrt{s}}$	$\frac{1}{\sqrt{\pi t}}$
$\frac{1}{s^{3/2}}$	$\frac{2}{\sqrt{t/\pi}}$
$\frac{1}{s^a}$	$\frac{t^{a-1}}{\Gamma(a)}$
$\frac{1}{(s+a)^k}(k>0)$	$\frac{1}{\Gamma(k)}t^{n-1}e^{-at}$
$\frac{1}{s+a}$	e^{-at}
$\frac{1}{(s+a)^2}$	te^{-at}
$\frac{1}{(s+a)^n}$	$\frac{1}{(n-1)!}t^{n-1}e^{-at}$ (n = 1, 2, 3…)
$\frac{n!}{(s+a)^{n+1}}$	$t^n e^{-at}$ (n = 1, 2, 3…)
$\frac{1}{s^2+\omega^2}$	$\frac{1}{\omega}\sin\omega t$
$\frac{s}{s^2+\omega^2}$	$\cos\omega t$
$\frac{\omega}{s^2+\omega^2}$	$\sin\omega t$
$\frac{s}{s^2-\omega^2}$	$\cosh\omega t$

6.8.2 Trend Analysis

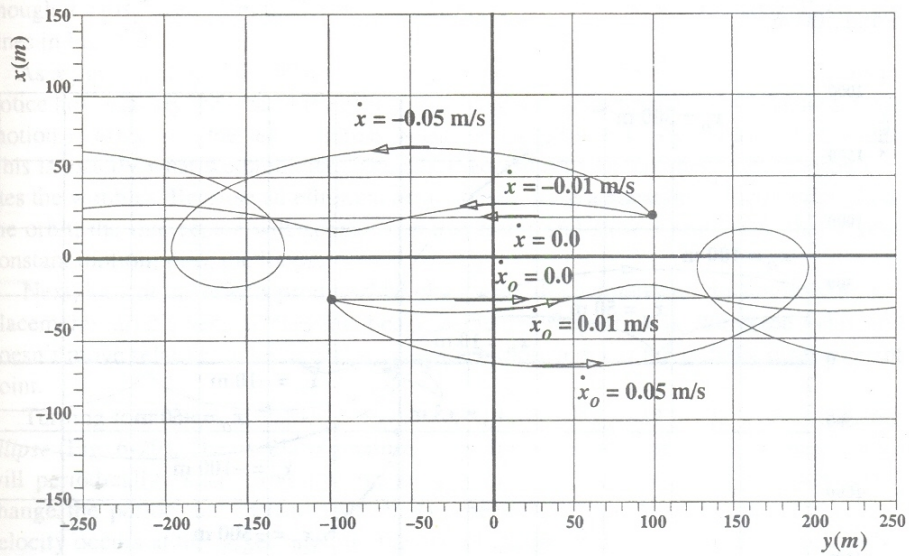


Figure 6-27. Motion of Interceptor for x_o and y_o Displacements, $\dot{x}_o$ fixed and $\dot{y}_o$ and $\dot{x}_o$ Variations. Varying the interceptor's initial conditions gives us several interesting motions including straight lines—$\dot{y}_o$ = 0.044 15 m/s for x_o negative, and $\dot{y}_o$ = −0.044 15 m/s for x_o positive.

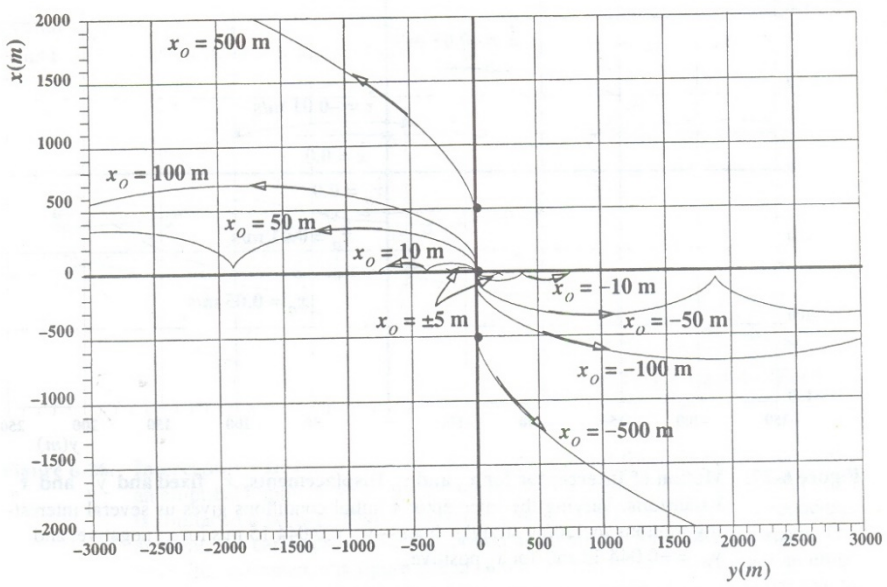


Figure 6-28. Motion of the Interceptor for Various x_o Displacements. Notice how all displacements above the satellite move to the left (lower velocity causes a lagging effect), whereas those below move to the right (higher velocity causes a lead). In addition, as the satellite's initial location gets farther from the target, the motion away from the target increases.

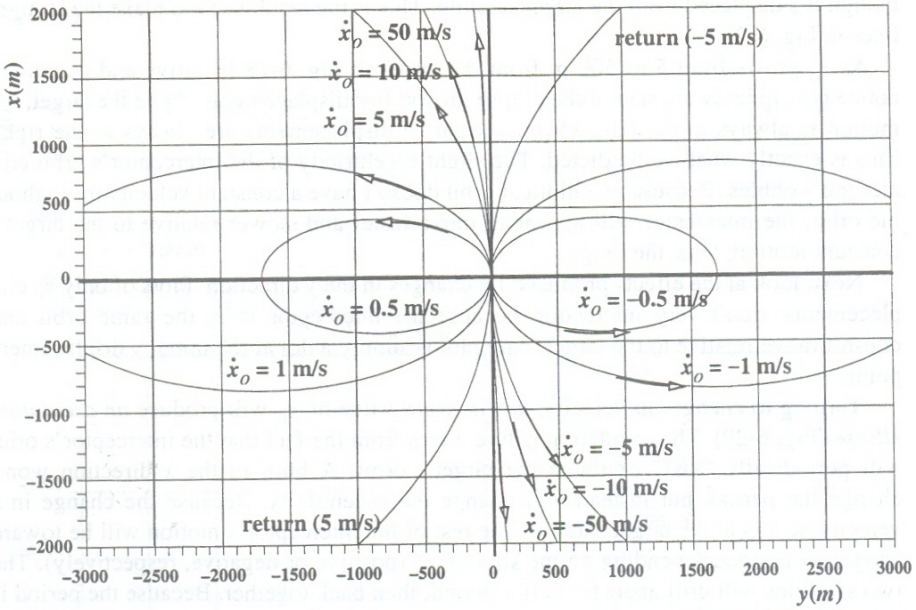


Figure 6-29. Motion of the Interceptor for $\dot{x}_o$ Variations. All motions which result from variations in the x velocity become periodic motions about the target. Notice each path *will* return to the target in one period.

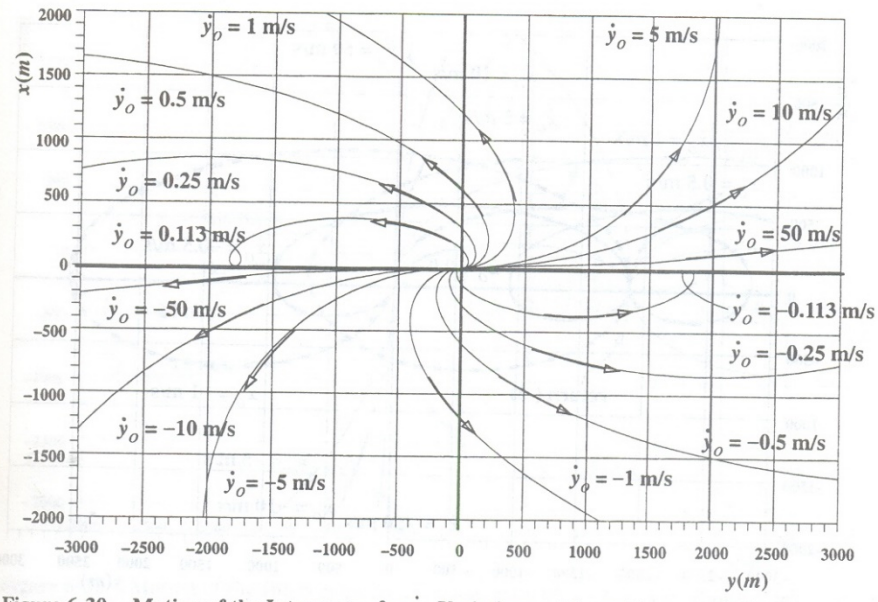


Figure 6-30. Motion of the Interceptor for $\dot{y}_o$ Variations. All positive y velocity changes cause the interceptor to travel left, whereas negative velocities cause motion to the right. Although $\dot{y}_o$ = 50 m/s appears to travel to the right, it will curve back around to the left over time.

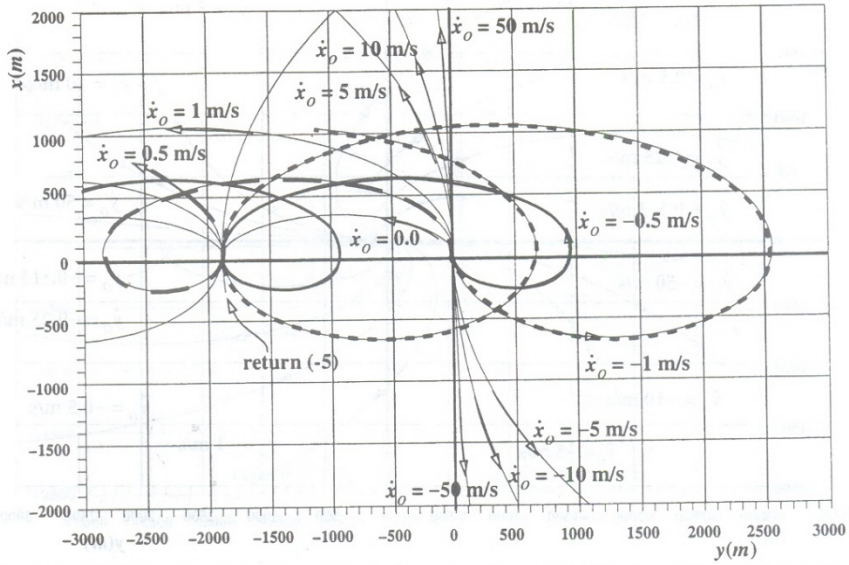


Figure 6-31. Motion of the Interceptor for x_o Displacements and $\dot{x}_o$ Variations. This figure represents the most complex motion of the individual changes. The graph is based on a constant offset of 50 m for x_o, with velocity changes in the x direction as indicated. Notice how many of the curves converge at the same point (near the return). Eventually, all paths will end up drifting behind (to the left).

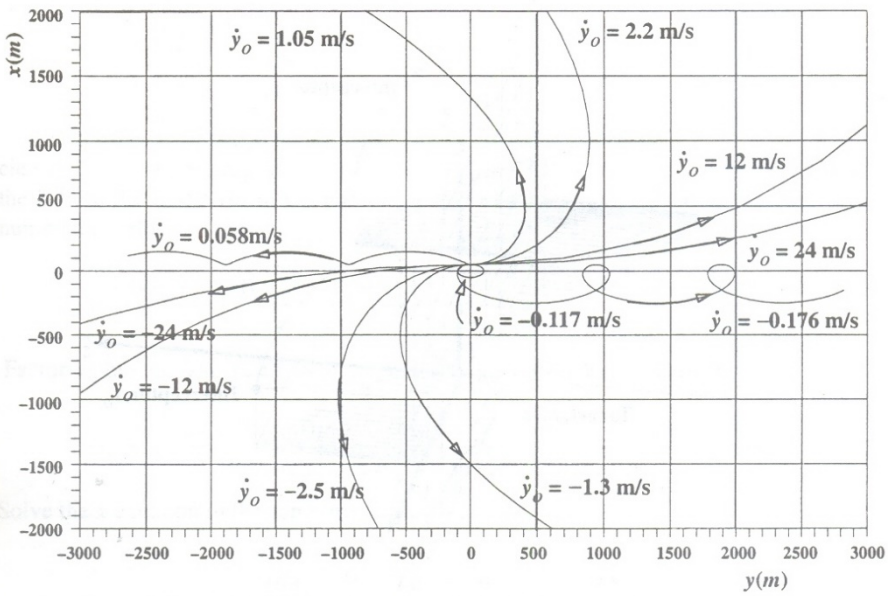


Figure 6-32. Motion of the Interceptor for x_o Displacements and $\dot{y}_o$ Variations. Each of the variations above comes from a 50 m x_o displacement coupled with varying values of y velocity as shown. Notice the special case of an "orbit" around the target satellite at –0.117 m/s.

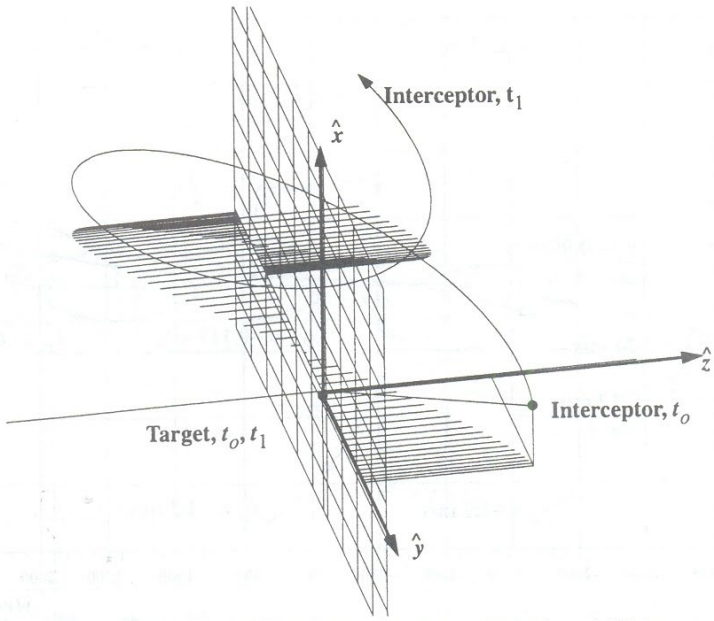


Figure 6-33. Relative Motion of the Interceptor with a *z* Variation. Adding a *z* variation to any of the other motions described simply adds a three-dimensional component. In this example, the interceptor goes above and then below the target satellite.

- Analytical Solution for Initial Velocity for Rendezvous/Docking

[x(t) y(t) z(t)] = [0 0 0] means that the interceptor has docked with the target satellite after time *t*

$$x(t)=4x_0+\frac{2\dot{y}_0}{\omega}+\frac{\dot{x}_0}{\omega}sin(\omega t)-(3x_0+\frac{2\dot{y}_0}{\omega})cos(\omega t)=0$$

$$y(t)=\frac{2\dot{x}_0}{\omega}cos(\omega t)+\left(6x_0+\frac{4\dot{y}_0}{\omega}\right)sin(\omega t)-(6\omega x_0+3\dot{y}_0)t-\frac{2\dot{x}_0}{\omega}+y_0=0$$

$$z(t)=z_0cos(\omega t)+\frac{\dot{z}_0}{\omega}sin(\omega t)=0$$

Substitute

$$A=\frac{4}{\omega}sin(\omega t)-3t,\quad B=\frac{2}{\omega}cos(\omega t)-\frac{2}{\omega},\qquad C=y_0+6x_0sin(\omega t)-6\omega x_0t$$

$$D=\frac{sin(\omega t)}{\omega},\qquad E=4x_0-3x_0cos(\omega t)$$

into the above equations results in

$$-B\dot{y}_0+D\dot{x}_0+E=0$$

$$A\dot{y}_0+B\dot{x}_0+C=0$$

Multiply the first equation by (-B/D) and add to the second equation leads to

$$\left(\frac{B^2}{D}+A\right)\dot{y}_0+\left(-\frac{BE}{D}+C\right)=0$$

$$\Rightarrow\ \dot{y}_0=\frac{BE-CD}{B^2+AD}$$

The first equation is rearranged as

$$-B\dot{y}_0+D\dot{x}_0+E=0$$

$$\Rightarrow\ \dot{x}_0=-\frac{E-B\dot{y}_0}{D}$$

Now re-substitute the definitions of *(A, B, C, D, E)* into the above planar velocity components:

$$\dot{y}_0=\frac{\left[6x_0(\omega t-\sin(\omega t))-y_0\right]\omega\sin(\omega t)-2\omega x_0\left(4-3\cos(\omega t))(1-\cos(\omega t)\right)}{\left(4\sin(\omega t)-3\omega t\right)\sin(\omega t)+4\left(1-\cos(\omega t)\right)^2}$$

$$\dot{x}_0=-\frac{\omega x_0\left(4-3\cos(\omega t)\right)+2\left(1-\cos(\omega t)\right)\dot{y}_0}{\sin(\omega t)}$$

Since the *z*-equation is decoupled from the *x*- and *y*- equations, we can find $\dot{z}_0$ individually

$$\dot{z}_0=-z_0\omega COT(\omega t)$$

The above analytical expressions for $(\dot{x}_0,\dot{y}_0,\dot{z}_0)$ enables us to calculate the initial velocity of interceptor S/C for a given initial position (x_0,y_0,z_0) to intercept the target S/C after time t. Be sure to calculate $\dot{y}_0$ first because it is needed to calculate $\dot{x}_0$.

For a rendezvous, interceptor S/C should perform the second maneuver to reduce its relative velocity to zero when interceptor S/C reaches relative position (0,0,0).